

ActInf GuestStream #018.1 ~ Michael Kirchhoff & Julian Kiverstein

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hello everyone welcome to actin flab guest stream number 18.1 we're here with julian keverstein and michael kirchoff as well as dave dean and steven from actin flab and we're really appreciative for the authors for joining today to be discussing some of their work we're going to have a presentation by the authors and then following the presentation we'll have ample time for discussion so please feel free to write any questions down in the live chat and we'll get to it so without further ado um julian and michael thanks a lot for joining and please take it away thanks so much for inviting us we've been we've been looking forward to this particular session and thanks for staying up late in the us and whoever you are in europe for getting up early it's only 3 30 in the afternoon here for us so that's more manageable in australia i'll just share my screen let's see and share is that right yeah you guys can see that bring that up so that should all be clear in a second is that right looks good great um so the paper is called the literalist fallacy and the free energy principle um it's a it's a philosophical paper on the topic of scientific realism scientific antiviralism instrumentalism in the context of the fep or the free energy principle um i suppose just as a primer this is a revised and resubmit stage for our paper at this particular moment in time so any comments that we might be able to get to further improve the quality of the paper will of course be welcomed so let's let's start there will be a little bit of i suppose conceptual stage setting at the start just given the nature of this particular paper and also because it sort of speaks to a few other issues that go beyond just a sort of narrow focus on the free energy so this is the debate as it as it sits or at least as we try to sketch it in the paper between scientific realism and instrumentalism so let's just say briefly what scientific realism is and briefly what instrumentalism is and then i'll we'll say a little bit more about how to think about scientific realism overall the point of the paper was to sort of try to marshal a set of arguments for why at least a version of scientific realism can be accommodated by fep models and it was motivated in part because it seems like in the last few years there's been a there's been a flurry of papers sort of in one way or another arguing for

scientific anti-realism with respect to the fep or how to interpret parts of that particular model so that's the kind of that's the kind of setting we're in so scientific realism that's the view that one reasonable goal for currently best scientific theories scientific models is to offer literally true or probably true or approximately true or probably approximately true descriptions and explanations of target systems i have here in bracket for example natural phenomena but it could also be artificial systems in the context of artificial intelligence artificial life for example now by contrast then instrumentalism or scientific anti-realism that's the view that scientific theories and models are nothing but instruments for prediction of observable behavior of target systems so here the the claim is something like this um from an instrumentalist perspective theory slash models they have explanatory power and that gets them some utility but they're not in any in any sense of the word truth producing so that one is not to think about instrumentalists or the instrumentalist position as uh putting forth statements about how the world is on the basis of their models by which you can say something say true or false about how the world is so the main question now is for this particular paper we are considering is the fep truth producing in a way that's consistent with scientific realism now one of course as i mentioned just a minute or so ago we think it is indeed consistent with a particular way of conceiving of scientific realism maybe a footnote here that's important both scientific realism and instrumentalism are quite a multiple there's multiple different versions of each particular uh framework and that was it science so we're trying to offer a reasonable general characterization of these particular frameworks in order to sort of get some traction on the debate so as i said there's a reasonable goal of our best scientific theories and models to offer these true or probably true approximately two descriptions of now here we we we draw inspiration from peter guthrie smith uh his book here erian reality that tells us how perhaps to interpret that particular general claim he says the following he says it's a mistake to express the scientific realist position in a way that depends on the accuracy of our current scientific theories so if we express scientific realism by asserting the real existence of entities recognized by science at this particular point in time then if those particular theories turn out to be false well then the whole framework of scientific realism turn out to be false so we shouldn't think that's what you have to do in order to advocate for a kind of scientific realism about scientific models theories and their relation to target systems so building on this we can say so again this notion here is that an actual unreasonable aim of science is to give us these accurate descriptions well that's not the claim that scientific theories must be true now and make truth producing claims about target systems now i mean it's a nice thing if they do but it

is not a necessary condition for for one to adopt scientific realism about scientific theories models there's no commitment to the success of the individual sciences so there can there can be trouble in biology but we can still hold for instance uh scientific realism about for instance the fep applied to the cognitive sciences for example now importantly and this is and this is why i'm just spending a little bit of time initially on scientific realism the kind of version of scientific realism we're interested in is going to allow for theories and models to posit idealization for instance understood as distortions approximations understood here is perhaps inexactness of scientific models they can propose fictional entities for instance entities that are not actual and thus like in the here and now but the important point is that you can keep that view and still endorse scientific realism because one aspect here is that the long term aim of working with your scientific models is for them to become truth-producing or to become more and more aligned with the target systems that we're interested and we'll finesse this viewer sort of move along now this issue between the scientific realist and the instrumentalist as we are considering it in the context of the fep that comes up when we consider what we call the map problem and this is not a term of art that we have invented the other authors in the both in the fep and elsewhere that make use of this for instance mel andrews in her nice paper from 2021 in biology and philosophy that speaks of this particular problem so this is just a quote from the paper that gets thinking about what our focus is so we're going to ask what is the relationship between scientific models constructing using the fep and the realities these models purport to represent our focus is especially on the fep and what if anything it tells us about the system that is used to model and that this is what we call the map problem how does the map your theories your models relate to the territory real world target systems of which those theories and models are maps now if there's anybody joining on the live stream um this is just a very brief statement about what the free energy principle says julian and i and as well as ian are working under the assumption that that doesn't need to take long to introduce that in this particular context but very briefly then the ftp states that a self-organizing system that tends towards maintaining a non-equilibrium steady state with its environment must minimize free energy where minimizing free energy at least under a certain invasion interpretation is to maximize model evidence that is maximize the likelihood of some data even a model of the system's interactions with its environment now how does the map problem arise in the context of the fep well it might arise if you're a say a researcher working in artificial intelligence and the aim is to build robots that engage in active inference for example navigating or making decisions etc so here the question becomes what's the relationship between our fep model

understood as an active inference model and then how one engineers if you like artificial agents to perform in accordance with those particular models what those models specifies in terms of say decision making belief updating and selecting action policies for adaptive behavior a different way the map problem might arise would be from the computational neuroscientific perspective where one might be interested in tracking transitions across belief perhaps in deep generative models and the map problem then is well how does that theoretical specification of free energy minimization under active inference how does that map onto what the brain actually does in order to say shift from beliefs to beliefs or to make decisions about what to do in the world so that's just a few examples of where this particular issue becomes prevalent now just to situate this discussion a little bit because there's a few tweaks and turns in the paper and it's a large and somewhat complex conceptual landscape we're working in or at least it seems to me um we're just gonna try to situate this a little bit more so you get a feel for this um we got a nice reviewer comment uh that sort of introduces the issue here so the comment is something like if we take the free energy principle like many other theories across different sciences then we can think about that as a kind of model based approach to science and if you look at the little image over here which is a modification of ears uh sort of a little description of how this works modified by graffiti smith in the 2009 paper we can initially start to develop a description of a model and to say something about what the model system in question is and the comment is something like this we can study models for their own sake as andrews has pointed out male as i referred to earlier any model that can be treated realistically can also be treated instrumentally scientists can disagree about which bits of the model are meant to be real now this is important in understanding how the map problem arises so andrew's in the 2021 paper gives us a whole list of different ways one can see the kind of mathematical structure of the fep being interpreted as a scientific model the work of michael weisberg is influential in the context of philosophy of science philosophy of biology so she and he they can with sort of a minimally three different models that we're interested in the fab essay sort of target less model the ftp is a generalized model and then the fep is a target directed model now you can work purely on the mathematics if you like of the free for the sake of just understanding the model for getting your model description right of your model system as per that image now here the point is that that in and of itself doesn't give rise to anything like the map problem because you can disagree you can come at that that work from an instrumentalist perspective you can come at it from a scientific realist perspective but doing that kind of work doesn't really allow you to adjudicate between the two positions or so it

seems to us so that would be the instance of a target-less model you're not asking questions that relate from the model to a target system you can treat the fpp as a generalized model you can perhaps think of it as you could model theories natural selection from the point of view of the fep but we are not saying anything about the evolvement of any particular kind of species that maps on to for instance a different discussion as well which we consider but we won't talk to unless it comes up in the q a about the role of the markov blanket notation now as we put it in the paper that gives you a kind of generalized approach for delineating boundaries but it doesn't give you a approach for delineating boundaries for any particular individual in that sense it's a generalist take on fep models on the basis of the markov magnetic formulation now the map problem first gets interesting when you pursue a target directed model when you try to implement the mathematical structures positive by the fep in terms of a process theory such as active inference so you're sort of moving from a mathematically general characterization of the adaptive behavior to specifying what a particular individual or a set of individuals must do in order to engage adaptively in the world so you start committing yourself to an act to to getting the sort of the right-hand side of this this model uh or this little image if you like uh saying something about in which way do your model now resemble the systems that you're actually aiming to explain or understand something about okay so we should so once we have that we should give the argument for the instrumentalists and then once we've done that we'll see the kind of problem that we raise for the instrumentalists that we think is a kind of a fallacy of inference from certain considerations to certain claims about the feb so premise one here is that active inference models on the under the fep introduce distortions in representing biological and cognitive systems via idealizations so one way to think about that might be the introduction of variation of free energy as a kind of distortion into the framework in order to explain you know it's kind of solve a computational problem about how surprising it's minimized as well as in exactness via approximate and one example might be something like introducing this idea that free energy minimization can be construed mathematically as akin to a kind of bayesian inference would be an inexactness introduced into the formalism primarily because well so it seems nobody thinks that the brain quite literally encodes something like basis theorem okay premise two then is that scientific realism requires that models provide descriptions with target systems that are literally true so that's a somewhat of a denial if you like uh if you were the whole scientific realism with respect to premise one premise three then is that active inference models are not true and accurate representations of biological and cognitive systems precisely because of the issues raised in premise one therefore active

inference models are false there are best useful fictions and just to give you the references of where we are drawing work from in terms of defining the term idealization and the term approximation here we're using a nice paper by mcmullen on galilean idealizations to think about idealization and we are using a canonical paper by norton to define approximation these are somewhat contentious primarily because there's as one might expect there's all kinds of other takes on it but you have to come down on one side i suppose so once we have the instrumentalist claim up and running this is the kind of fallacy that we're arguing that a set of set of papers defending instrumentalism in the context of the fep is making and we coined the literalist fallacy and just the prime use though premise two here is the claim that scientific realism requires that models provide descriptions of target systems that are literally true hence the claim the literalist fallacy because we're going to show that's that's not a necessary condition upon which the base one's defense of scientific realism so what we say is the mistake the literalist fallacy is the mistake of accepting or affirming about the fep from the fact that active influence model cannot literally map onto a slightly different formulation that we have here is that in this context of the fep some find apparent discrepancies between the map and the territory a compelling reason to defend instrumentalism about the fep and we take that to be problematic given this particular kind of fallacy we identify this particular fallacy by those defending instrumentalism about the fep and we call it the little fallacy here's this the second variation of it if you like this is the fallacy of inferring the truth of instrumentalism based on the claim that the properties of fep models do not literally map onto real world target systems so so one one question that might be interesting to discuss in the q a is whether or not it is indeed the problem for a instrumentalist to infer the truth of their own position on the basis of the claim that the properties of fep models do not map literally map onto real world so the pushback here on the second formulation is something like this instrumentalists are not asserting the truth of their own position that would be self-contradictory now there's a set of conceptual issues here but i just wanted to flag the issue and perhaps come back to it in the q a it would take us too long to get into it and they did somewhat philosophically conceptual here's some here's some evidence that you like uh a set of papers across some different topics that that we arguing are all committing this this kind of fallacy is a 2020 paper by venice and politico that state they said that it remains quote it remains disputed whether it's the fep statistical models or scientific tools to describe non-equilibrium steady-state systems which we call the instrumentalist reading or are literally implemented and utilized by those systems the so-called realist reading and what they're going to conclude is that since fep models are not

true and accurate descriptions of these target systems instrumentalism turns out to be the only option so for us that's a bit that's the literalist fallacy because it assumes that the realist decision is that your theoretical models literally have to be implemented in target systems for that kind of framework and the philosophy of science to be up and running we're in about adele 2021 presumably updated 2022 at some point this year in their target article in bbs focused on the ontological status of the markov blanket formalism in the fep now they argue that much of the literature on the fep implies that organisms literally instantiate the mathematical structure of markov blankets they argue that such a use of the formalism conflates a model with its target system now if you think that in order to use the markov blanket formalism productively even from the scientific realist point of view that the mathematical formulation of that particular formula has to be implemented in the actual target system then you're permitting the literalist fallacy because even the scientific realist is going to push back against colombo and palacios in a 2021 article in biology and philosophy target the issue of herbioticity in the fep namely that one can realistically model biological systems as having an ergodic density now they deny that ergodicity captures properties of biological systems and therefore the fep is biologically implausible now the line on that is that's that's a kind or a species of the literalist fallacy precisely because it's going to assume that the theorists actually think that it's part and parcel of target systems that they have an ergodic density our take on it is no that particular approach to modeling uh state transitions uh under the fep is a precisely that it's it's a kind of um it's a problem it's a sort of a it's an scientific convenience in order to say something about how systems occupy states over time but you shouldn't conflate that particular uh perspective with saying that's literally the case of target systems i'll skip the last quote here just for time considerations we can come back to if there's there's any issues okay let's turn to some examples and some cases of this literalist fallacy and how to how to think about it than in terms of scientific realism so in the paper we consider a bunch of these sort of cases or examples we consider the notion of variational free energy we consider the markov blanket formalism we consider the role of bayesian inference in these sorts of formulations and then finally we turn to consider the topic of ergodicity now for this particular uh presentation if you like we're only going to focus on the first variation of free energy and then the latter epistemic and perhaps we can come back to the other cases in q a it strikes us there's two different ways of interpreting the variational free energy notation primed here under the consideration that it seems as if organisms are minimizing variational free energy in the long run so the as if formulation here which seems ever pervasive now in the literature um is not i think a one-way

street there's a couple of options here that we now want to try to unpack especially because we're doing the presentation and especially because we've been prompted to do so i think by by a set of revenue comments but let's get into it this is the sort of first option if you like that's gonna that's gonna deliver a set of reasons for thinking that instrumentalism is the case for the fep and we're gonna push back on those particular reasons so uh variational free energy is an information theoretic construction used to compute how organisms are able to resist decay by positing a variation of free energy in an aquaponic rival now under the fep for an organism to exist it must keep its states within certain bounds okay so that's the generic opener so i'll give you some examples now of the as if formulation uh sort of prompted by this quote from fristen the fep is a mathematical formulation of how adaptive systems is natural to decay ramps that and colleagues 2020 paper say this means that internal and active states will look as if they're trying to minimize the same quantity namely the surprise law of states it constitutes the thing a particle or some preacher so here we have the as if statement they appear to perform this this kind of minimization of variation of free energy similarly in a 2019 paper um any system that avoids surprising exchanges with the world will look as if it's predicting tracking and minimizing a quantity called variational free energy on average and over time now i've included this because we're going to speak to this one this particular paper as a second option for how to think of the acid formulation and how perhaps the square acid formulations with a kind of then as 2020 the system does not actually predict infer track or minimize the quantity called variational free energy but it merely looks as if the probabilistic model merely tracks certain real statistical relations in the organism environment system so this particular reading here this first reading the as if formulation of variation of free energy leads to scientific realism about active inference models under the fep and it does so essentially because we have to sort of treat systems only as if but not actually minimizing this particular quantity yeah just jump in on this a minute so the ramstead quotes there are saying that uh what it looks as if the system is minimizing is surprisal whereas van s shifts to talking about variational free energy so um i think we could accept that it looks as if the system is minimizing surprisal because variational free energy is this bound on surprisal yes so the ramstead quotes are actually consistent with yes that they could be read as allowing the system really does minimize variational free energy and it looks as if in doing so it's minimizing surprisal so perhaps we can come back to that in the discussion yeah i don't think that ramsteid and colleagues are necessarily best read as endorsing instrumentalism i think it's quite consistent with realism yeah no i agree and i i think we'll see that come out i sort of recently explicit when we turn to

the second option to just step in if i don't if i don't actually manage to pull that out uh explicitly very good yeah basically what we can now assume is that variational free energy under this first half reading is a mathematical fiction at least that's what we're told these sorts of constructs are used for fictions uh but they do not actually exist so how to kind of think about that then because we we tend to agree with that particular point you can think that variational free energy a bit like a galilean frictionless plane for example is a concrete entity as in it has some mathematical concreteness about it but it's not one that actually exists in target systems here the mass is not the territory so even though we can we can we can kind of think of friction as planes as being concrete in one sort of sense of its mathematical concreteness one wouldn't want to infer that that julian is a frictional plane for example that that would be that would be false and it would ultimately end up in in a bad set of inferences now here again we draw on the work of guthrie smith in order to think sort of more generally about the use of fictional entities in the context of science more generally but here's what he says generally speaking he doesn't dress anything to do with the fep just to make that clear now fictions do not exist he says but at least many of them might have existed and if they had they would have been concrete physical things located in space and time and engaging in causal relations so it's kind of counterfactual test you can run on fictional entities and science from the point of view given here now you can say fictional entities such as variation or free energy that might lack actual existence and can be useful for understanding target systems nevertheless so you might say that active inference models provide insight into the workings of biological systems for instance with helping to understand what a system has to do in order to maintain homeostasis you can also say something like the following meaning that under a kind of basian interpretation of how one minimizes this particular quantity you can draw some similarities to how that to how um surprisal is minimized in actual target systems by understanding something about the minimization of variational free energy but there's a kind of similarity relation one can draw between depositing or fictional entities the theoretical models and then what actual target systems are doing just to give you a sense of this so motivated by active inference models on the dfep par and colleagues say the following about the significance of their work on active inference i'll just give you the quote and you can kind of see this this is coming in to to to being so in brief we use magno in circular graphic graphy in combination with eye tracking to assess the neural correlates a form of short term memory doing a dot uh cancellation task using dynamic remodeling to quantify changes in effective connect connectivity we found evidence that the coupling between the dorsal and ventral attention

networks changed during the psychotic interrogation of a simple visual scene now they say here and i've highlighted this intuitively this is consistent with the idea that these neuronal connections may encode beliefs about what i would see if i were to look there and that this mapping is optimized new data are obtained with each fixation so the idea here is that fictional entities such as variational free energy allows for the further understanding of target systems by making precise how target systems may minimize free energy whether or not that be variational free energy or expected free energy in the sense of minimizing their surprisal and not directly in the sense of minimizing these two notations so the summary here that we give in the article is that the fep comprises what is known as indirect representation of a target system so it says something indirectly about the operations of target systems even if it involves the use of abstract entities it's an idealized approach representing complex or unknown processes in the world it's standard practice to view scientific models as indirect representations of the real world target system now furthermore most theoretical models are composed of a set of mixed claims now this is going this is rather important for us to sort of preserve a notion of scientific realism about the fdp basically and we use the work of spiel out here this means that the model will pass it if through the presence of what he calls both okay entities such as electrons and their ilk and supposedly non-okay entities such as numbers or theoretical ideals now we think the fvp is something like such a model it's a model posed or composed of mixed claims it put forward all kinds of okay entities neurons reflex acts etc and perhaps what you might think of as non-ok entities such as the variational free energy notation where non-okay entities has to be understood in terms of not being literally true of their so this gets us to the second formulation that julian alludes to alluded to earlier of the asif formulation now take us an example here from dennett's 1996 book kinds of minds and i think thanks to an anonymous reviewer here for pointing us to this this idea then it's so-called screeches now these creatures are said to be able to select from possible action policies although it's not put exactly in those in those terms in order to prune away inferior options to avoid fatal consequences so in this sense these creatures have some kind of inner environment where they can select amongst possible actions now here's the tales of two densities paper that i co-authored with max ramster and cal fistin i'm not going to read you the quote here but the basic idea is going to be something like this that one can prune away unwanted action policies by testing what kind of expected sensory observations one might encounter will want to elicit certain kinds of actions but whether or not one prunes away action policies the crucial part here is that that kind of beliefs transitioning over possible sensory outcomes given certain

kinds of actions can itself be characterized in a fictional way in in in a form of fictionalism as julia mentioned earlier that is entirely consistent as far as we can tell um with scientific realism so for the puberian active inference feature one can think of trajectories and suit over possible action policies in terms of fictions as i just mentioned and now we come back to our our formulations of a fictional entities and science from a previous from a couple of slides ago now you might say that action policies are concrete belief states but they're not actual they do not actually exist they do not actually exist given their counterfactual hypothetical characteristics yet whether to exist they would be actual with some causal powers there will be implications for inferring one over the other arguably some of the hypotheses come to fruition and therefore exist so what we have here is an sf formulation of the of fep explanations has a fictional characteristic to it so part of the entities that compose our fep explanations are fictional entities but they're entirely consistent with scientific realism about fep explanations given the considerations about the status of fictional entities and science in general okay so let's see we've been going for about yeah 35 minutes i'll start wrapping this up so we can get to the q a second case one slide really quickly about ergodicity just to just get us thinking about this at least we can come back to it now fristen defines ergodicity in the following way one can interpret the average amount of time a state is occupied as the probability of a system being in that state when it's observed at random okay now colombo and palacios in this 2021 paper in biology and philosophy they say that phase spaces of biological systems are structurally unstable they're always changing and unpredictable unpredictable making them non-ergotic there isn't a set of fixed states if you like that the system keeps returning to if such a system is a biological system on this particular account by columbia and palacios now they say that the fep trades off realism for generality and mathematical precision and they do so in a sense to to to sort of put pressure on this idea that the feb can be thought of in terms that are consistent with scientific realism now it's true if you have you take a the approach to the fep as a generalized strategy to modeling that the fep trades off realism for generality and in the same time maximizing mathematical precision but i think here's here's the twist so this would be the first way of responding to this particular claim if you turn the fep the mathematical structure into a target directed system what we get here we get to work with specific systems so now you get to increase the level of realism in your in the relationship if you like between your model and your target system a second way of thinking about ergodicity is that yeah okay it be true that they don't map up they don't have biological plausibility but you can think of ergodicity as a convenient modeling assumption so the fap doesn't claim that biological

systems are literally ergodic there's even the loosening in the literature now to saying locally ergodic or there's even a further loosening the shifting talk of ergodicity uh entirely to talk about kind of nest densities and so on we where you can actually still retain this kind of metastability this this level of change and this degree of unpredictability so to conclude arguments for instrumentalism about the fep at least the ones that we are considering i'm sure there's one or two we have not considered trade on the literalist fallacy they accept or affirm instrumentalism on the grounds that the properties of active inference models under the fep do not literally map onto biological systems if we are correct then there is a version of scientific realism that is a live and tenable option with respect to efep thank you awesome great times um perhaps for the first word we'll go to julian just feel preach maybe say hi give your first take and then we'll go to the acting flab participants yeah so thanks for having us it's going to be fun to talk about this paper um i think we should just jump straight into discussion rather than me add anything onto what michael already said he gave an excellent summary of the paper so let's see what you guys make of it okay so age or beauty where do we begin uh dave or dean or stephen who would like to go first all right stephen first and then either dave or well i'll just like say thanks for the book that you published a few years ago about the third wave it was actually the key book that kind of made me keep going with active inference after i first heard about it because uh so it was a really really cool supposed to hear boundaries because i've been looking at inactive embodied approaches and so just that's the first piece um and um so in there you talk about the idea of markov blankets you're not directly but the idea of things being dynamic and changing and um i'm curious whether when you're thinking about instrumentalism there's a there's a in some ways the the danger of instrumentalism or is it starts to everything gets referred back to maths and then the maths isn't tethered to reality because it's instrumental and you can't it can be kind of a challenge then as we try where we're trying to bring in the four e's um to think about well how how does this start to map onto the way that organisms are trying to get a grip on the world in a bit more of a literal sense outside of the model and so um yeah i'd be interested in your thoughts about this this question of um the the ability for um the modeling to be used in something almost a way to get into phenomenon that might be happening during this minimization process michael do you want to start i i i can start um well one i think one thing that's important there just stepping back and thanks for the nice words on on the book that's always a pleasure to hear the instrumentalist i suppose is is just sort of as i said making a truth claim is off the table for an instrumentalist partly because the definition of instrumentalism is the sort of not to put forward any kinds of

statements about the world that can be true or false now they can say that you have you know you take in observations about how systems work and that's fine one one line there that's been pushed a bit in the literature now is to say well you can then you can be agnostic about whether those kinds of observations can sort of truthfully reflect or you know oppositely falsely reflect anything about the actual world now my sense is that and and this is just my sense that scientists given different tasks and activities questions and focus points will shift a bit between these and and that's in a sense why i had this notion of why we have this notion of of a targetless model i think precisely because that's the domain we we don't need to come down on one or the other side clearly the model for the sake of its for for gaining more understanding you know a nice example i was talking to an ecologist on the weekend is just thinking about these massive climate change models right with this hyper practical goal but yet they're so wonderfully complex that one spends one's career figuring out how the model works and that's and that's okay uh in a sense but i then shifting it and trying to answer uh your question i think some of the work that julian and i have been doing on the sort of ever or on occasion this is soften that shifting boundaries delineated by appeal to markov blankets tells us something concretely about such systems so we can start thinking a bit about the differences between say the spider and the spider web relative to with whether or not the australian or private has just eaten its web or whether it's producing its web so in that sense you might think that you know it's it's 7th sensory is shrinking and expanding depending on you know what time of day it is so yeah i might as and to add to that i i i'd like to think that but i'd like to think that it's the ambition of even theoretical neurobiologists or if you know from depending on the domain or issue they're targeting is to if you're working in the context of psychiatry on the basis of fep models do you want to understand something well actual uh psychiatric disorders that we as human beings may may go through and then and you want to understand the the mechanisms of how that of how that works so you can intervene in the right kind of way now if you want to do that presumably you have to say something about how those mechanisms work and so you can you can quite literally as i said manipulate them if that's possible in one when one way or another anyway so that's my initial take on that i want to go in a slightly different direction on the how does the free energy principle relate to these ideas about embodied inactive extended cognition well thomas van s and uh innis hippolyto both argue for an instrumentalist reading of the free energy principle in order to uh to show how it could be consistent with a non-representational understanding of cognition and so they are they're keen on avoiding the idea that the brain literally represents the world by encoding a generative

model and the way that they avoid that interpretation of the free energy principle is by endorsing the claim that the free energy principle is just a useful fiction for modeling the brain and we think there's a couple of different issues that are getting mixed up here so one is representational readings of the free energy principle and uh karl kristen often says that uh the brain um doesn't have a model but is a model that the organism as a whole is a model of its environment and here he's picking up on ideas from cybernetics like the the good regulator theorem the law of requisite uh variety is it from ashby [Music] that idea of the organism being a model of its environment we think can be read in a realist way but but also as suggesting a a non-representationalism uh so this came up a little bit when uh michael was talking about pop arian creatures where they're able to because of the uh hierarchical nature of the generative model and try out possibilities before they act on them and that's sometimes taken to imply in the in the philosophical literature the organism if it's able to to do that kind of counterfactual inference it must have some kind of model of these possible states of affairs that it's carrying out inference over and so that gets you back into a kind of um representational reading of the free energy principle again but uh we are offering uh an interpretation of the free energy principle where we take very seriously this idea that the organism is a model but doesn't have a model and therefore that opens up the space for a realist interpretation of the free energy principle um that is also non-representationalist or inactive and that's that i think the space that thomas fan s for instance thinks is not available to proponents of the free but rather than go into why we think he's wrong let's let me just stop there because i've said quite a lot already thank you for the responses dean well i think i'll ask both gentlemen for your opinion on this i i think if if you set your argument up as a literal one you're gonna you're gonna find yourself into trouble because i i think if this thing kind of gets resolved there's no debate if it's metaphorical because you have to have a minimum of two i mean with these markov blankets you got internal and external you need you need minimum of two just so that you have discrimination just so that you can tell variation in the world and that's what metaphors allow for so i'm just wondering why we're not arguing for both that it's the synthesis of the instrumentalist and that way to measure and filter what's in and what's out because we do measure that and and filter for that and also the relational piece i still like to bump into edges so i agree with you as an organism i am the model until i fall off the cliff right so if we if we looked at this as a metaphor instead of literally wouldn't that kind at least smooth out some of the differences instead of versus its and this equals that in in a metaphorical sense what are your thoughts in that i don't think you can have both

instrumentalism and realism although max ramstead has said things along those lines that he wants to try to well then i'm with max best of both worlds yeah but uh but the difficulty is that these positions actually come into conflict contradict each other because one says that the free energy does not map onto anything in the world so we can't make any claims based on the free energy principle about how cognitive biological systems are organized it's literally just a modeling tool so it can help us to predict make predictions about the behavior that we can observe of biological or cognitive systems but but the free energy principle doesn't correspond to doesn't map onto anything in the behavior or the organization of biological and cognitive systems and that to me seems in conflict with the idea that the organism is a generative model if you say the organism is a generative model then that seems to me to commit you to the idea that the free energy principle is describing something about how biological cognitive systems are organized and therefore it's uh it's so it ought to be construed in realist terms not in instrumentalist terms now you said something else about metaphor there yeah science introduces all kinds of metaphors and i think that that introduces a kind distinct set of issues really from the instrumentalism and versus realism debate because we can still ask for how do we interpret those so no doubt it's true that science engages in metaphorical talk all the time but it is once you get into the question of how to interpret the metaphors that then the realism instrumentalism issue comes back again so can i just do one quick follow-up if i focus on the or the realism and not focus on the link with the with the two arrow ends the metaphor i i i absolutely appreciate what you're saying julian if i if i pick one camp with the other then i'm not focused on the metaphor but what if i'm just exclusively looking at that link as opposed to the the opposing camps can i i i'm very comfortable i'm i'm a contradiction i'm a human being right so i'm not sure why i have to pick a side i'm you haven't convinced me about that yet but i mean depends a bit about how you how you flesh out the term metaphor here right so if we went back to the uh that little image i had from gear you had you know your model description model system and some target system so they say for instance you pass metaphorical entities i suppose is that what you have in mind there by metaphor i'm talking about i'm talking about the process of the of the way to be able to link to discrete or discriminatory ideas or objects or because i want to be able to hold up the relationship and the measure of that relationship i don't want to say that i have to discard one in order to take up the other maybe it's useful to bring in the markov blanket here as a metaphor for the boundary so the blanket seems like a good metaphor for capturing the boundary of a of a particle or an organism right so then we can ask well what is the relation between that markov blanket concept where you can where

the markov blankets are nested and so you can find them at multiple scales of organization multiple temporal spatial scales what's the relationship between that nesting of markov blankets and the self-organizing systems that we're using them to model so i think that's where you get to when you try to unpack the metaphor there's a nested markov blankets is a metaphor and how do we then interpret that and for michael and i the way to interpret that metaphor is to think of self-organizing systems as literally having this well not literally that's vacuous use of words isn't it as uh to think that there is a nested organization within uh self-organizing systems that can be modeled and described indirectly using the mathematics of the free energy principle or using the formalism of markov blankets so using the bayesian causality it's yes on that note you use these particular constructs i think on the basis of what julian just said to actually guide your research in the sense of figuring out something about how your target systems are organized so the one way of casting this in the jargon we've been using that you distort you introduce a distortion that you think you need to take literally for instance the mathematics underlying this particular markov language of formalism but you use that in order to guide and direct your search now that's important because that's the kind of key to our particular way of thinking for us about scientific realism that we can have all these distortions and all these points of inexactness in our theoretical models however the ambition is to say something about the target system that's the that's where the sort of conversation between the instrumentalists at least a version of that the version of the scientific realist framework uh come apart now there's i should say there's there's a set of good reasons for thinking that the two positions and philosophy of science are mutually exclusive they can't be true at the same time and one is the positive of a set of metaphysical claims on behalf of the scientific realist it's typically the case that one is committed to the following there's some kind of mind independent reality okay now that can't be the commitment of the instrumentalist because you can't possibly because you have no you're not producing the truth about how the world the second metaphysical commitment is that we can somehow get a grip on uh unobservable causes of our observations but that's not an option for the instrumentalists they can say something about their observations of your empiricists for example you can't you can't start making inferences from that to unobservable causes of those observations but metaphysically there's something at stake when you have a discussion that goes that considers these expansive sort of broader considerations and positive science but those are metaphysical commitments and i suppose we can argue so to make it more concrete again going back to the nested markov blankets and the reason that we

we describe self-organizing systems in those terms is because of how the the faster processes are enslaved by slower processes so you see this this circular causality that's described in uh complex systems theory and there's there's a there's a dynamics that you can observe living self-organizing systems um that can be in exactly and approximately described using the formalism of markov blankets and and it's because of that similarity or resemblance between the the formal apparatus that we're using when we employ uh base nets and the modeling formalism of markov blankets it's because there's that kind of similarity or resemblance between the models that we're making as scientists and the dynamics of self-organizing systems that we think it makes sense to interpret that metaphor in realist terms does that make sense well yeah i again i wouldn't argue that that if you if you do place yourself in one camp you you reject what maybe the other side is arguing what i'm thinking of is the the statistical probability as a distribution allows for perforations meaning the metaphor al is i wouldn't say it's a bridge but it's kind of an escape patch to boundary cross and i mean quite literally if i use a i i get out of the constraint of of not having that other person share the idea unless we can arrive at some convergence around something that we both understand from the past in order to be able to project into that that new set of congruencies so i don't want to take up all of this conversation i just wanted to see i think that the issue here is really something to do with what philosophers do where we like to disagree and argue with each other right that works for some people depending on their personality and for other people it doesn't work so well they prefer to try and find agreement or to find a way of communicating with each other for us and as philosophers i think disagreeing and arguing is not to fail to communicate is that it's just what we do as philosophers and what that made me think about is um very few people are in the position of putting their name and a year and a version on an idea and being like the um avatar of the idea like godfrey smith 2009 longino 2013. here's what the idea is dean or anyone who might be interested in applying these ideas might have less of a qualm with shifting their regime of attention even within a sentence from one usage to another and it's um in the fine art and science of philosophy where some of these um minds become fleshed out and we can really see like how deep logic goes and that's like kind of like higher overtones in the symphony of thought and then you know dean's just fiddling away on the subway trying to trying to get whatever currency they get up there in canada um let me go to dave with a question um because you haven't gone yet and then uh stephen afterwards okay um coming at this whole question particularly the question of metaphor my background is i'm coming at this from general semantics through the cybernetics theory of learning teaching through george

lakoff's grounded metaphor theory and through mark saul's conscious ego so when you say metaphor dot dot dot i say well who's actually driving the analysis who is it that has to be appealed to and i would say look the the interpretation is done by the metaphoric structure why is that a good grounding because the fundamental grounding of the entire of every living control system the speed of updating the the question of how existentially imperative that you not violate your priors down through very delicate questions of mathematical formalisms it's all grounded in the decline of avoidance or approach and decline of pain versus pleasure so it's all grounded it's got a basis your entire structure of of inference of induction your whole induction is already ground and so don't worry about what the metaphor means in terms of theories ask what do the theories mean in terms of metaphor don't be shy about telling me i've gone off the deep end every last person i respect has been accused of going off the deep end i might i might i might like i might let julian take first dips here so i can sort of get myself into into my own headspace all right um i like what you how you describe metaphors and uh and the place that they have within the free energy principle where they're grounded in these perception action emotion processes and for fep that what everything is organized around is these imperatives to resist disintegration and so it makes sense to me that the metaphors we use in in communicating with each other and thinking are in the end gonna inherit there and be grounded in life processes that's something that we talk about in terms of the life-mind continuity thesis coming out of inactive ideas in philosophy the philosophy of biology uh so there's an interesting connection there between uh the embodied cognition ideas that lack often johnson are using and and the the four e ideas so i think that's what you were just describing but that comes up at this from a slightly different direction to to what we're concerned with in our paper where what we're looking at is say a matlab model that somebody might make of uh in simulating active vision like thomas paris done micah allen when he's looking at how to model the interactions between heart rate and fear perception and he's constructing like a a simulation and in silico model using the mathematics of the free energy to describe those kinds of interactions between intraceptive processes like monitoring of your heart rate and extraceptive perception say fearful images of faces making fearful expressions so that's a kind of in silico model that's been made there and what can we learn from those sorts of simulations about say interactions between heart rate and visual perception and that's where our our sorts of questions come up because you can think while the the in silico simulations that are being made they're just um instruments for making predictions about say heart rate how heart rate's going to interact with perception and the the simulation can be useful for um predicting the

kinds of data that we could measure but the simulation itself isn't gonna give us knowledge of how the intraceptive system interacts with and and we just say no that's wrong the simulation can actually tell you something important give you a kind of knowledge maybe in the long term so we have to look over time and look at all of the refinements that are going to happen in active inference models before we can actually arrive at something that is a true accurate description but in the meantime the kinds of approximations idealizations that we're operating with when we make these kinds of in silico simulations are still telling us something important about how uh how the mind is organized how living systems are organized okay so that's the sort of claim that we're making can you see that that's sort of a little bit a tangent from what you were asking about metaphor maybe it isn't you can explain why you see how you see them coming together i have probably uh stated my my point in an overly political way i was very excited i was really i am very excited about what you're doing um kind of like when muhammad the the the opponent finally lands a really solid blow on muhammad ali and elise is okay enough playing around let's let's hit that uh no i'm very sympathetic i invited professor solms to this discussion i certainly hope he's going to watch the film um no i'm i'm i'm very much on your side i'm just saying i think what the i think the the the embodied perspective the um the pre-galilean perspective where valence where value where teleology viewed as the foundation or at very minimum a co-equal foundation of the entire intellectual structure the entire research structure i think that is exactly correct um and i'm really pleased for instance a subtle british nuance perhaps that professor used the same uh professor christian uses the term teleology so often in such an an important sense i think that already by itself involves hitting back yeah but he always says he's deflating teleology doesn't he he loves this word deflation something that we can see in in physics as well and uh yeah michael found something very interesting in a in a youtube video from pristin about uh where he was responding to some presentations that he'd heard where he described describes how uh this variational free energy relates to to gibbs free energy michael did you want to come in on that yeah i mean this is definitely well well beyond sort of my my pay grade with social media probably with some people in in our group here and in our audience well we'll be here basically the basic gist of it is something like this that you can you can describe transitionings across belief states by appeal to variational free energy and that's nice that gives you a nice sort of computational modeling language to to speak about state transitions across different beliefs so so the so the claim is here and it claim is interesting because it allows for an even tighter grip between your fictional constructs and actual systems that you can give a formulation of those state

transitions by mathematically showing an equivalence relation across the variation of free and the gibbs free energy which then by definition allows you to show in energy spent per joule or something like that but it what it actually costs to change one's beliefs for example uh in the context of the fep now that's that's that's great if that's the case because that brings you even closer from your model from your model space to your target system space but that that's a bit like when somebody asks me and okinawa well how does the octopus mechanistically select its action policies well i'm not a marine biologist and i'm not a specialist in octopi i mean that that that's not for me as a philosopher to tell you and i have i'm gonna i'm gonna cheat out and provide the same sort of reason for not going any further than that but it's a good example i think um yeah the reason i brought up was because you were talking about teleology and and what kristen does in deflating that notion is actually bring it back to to something in physics as well like gibbs free energy and the free energy principle shows you how you can take something that looks like it's going back to aristotle but actually show how to bring it into contact with um with the mathematics that we now have for understanding self-organizing dynamical i mean it's i think it's just an important note there so even if the term teleology is being used in the literature one ought not to confuse it with the sort of notion of a final cause as per the aristotelian system right so so we have we have sort of a final cause and ultimate end point that one necessarily strives towards right but rather that self-organization is somehow inherently purposeful and to me that's what strikes me as right but it also destruction is a deflated notion of teleology yeah but there's there's lots more we could say on this before it's for another session isn't it it's a little bit of a tangent from what we're currently talking about true cool um stephen and then i'll have a question and then we can sort of land yeah i think this this question about what we can model with the fbp or what fep is able to be used to do modeling on using variational free energy the model is and i think you've kind of resolved this to some extent by distinguishing modeling from models so there's instrumental models there's realist models and there's this process of modelling and one of the things about non and i think in some ways you don't say non-linear i mean non-linear dynamical systems might be better than that the the thing with those systems is is that they are in this case anyway generative and we've been looking at and i was actually talking to daniels in our tools session like about how these bayesian graphs can be thought of and one of the things is that generally of a modern you try and fit things to your model like in psychology you fit someone to the model of psychology right that can be the problem because now this person's being fit to sit with all these models well in this case these models generally always generate the data and but

they generate a little bit different each time because it's stochastic so you see that the way they evolve and that's what we're sort of getting at so building the model can be a useful design thinking tool to think about how it would be structured and then you get this idea of modelling and and this is actually maybe timeless with what dave was saying a little bit because my belief is that at the moment all the ways that we are able to model are very low dimensional like extremely low dimensional however i do believe i've been looking at stuff where phenomenologically it would be possible to use experience as the raw instrument so i think high dimensionality can be done through there is a potential for high dimensionality so there's this question of like when we're modeling using models which are low dimensional to such an extent and they even like often are partially observable markov decision processes so they're not even actually saying the data coming in has been aching necessarily through you know you either have these psychological models which are very low dimensional and are partially observable so it's gone up and it's hit the point of beliefs and then they start to look or they have these incredibly particulate models starting from like the first atoms of life or whatever i think that your intuition is probably right actually in terms of there are because i've we've been looking at some of this in terms of phenomenologically there is there ways that experiential information can be modeled in or put into these models and so it doesn't only have to be these very low dimensional ones which may be so low dimensional that maybe that's part of the problem it's not that the fep real it's just what they're using to model it is still so low dimensional yeah that's a nice question um let me at least attempt a form of response um so julian and i ian met the other day and was just sort of chit chatting here and since you did mention micro precision processes i suppose one way of conceiving of this if you like low dimensional versus high dimension or kind of simplicity versus complexity but kind of inaccuracy versus more accuracy distinction would be just a sort of quite often one finds that those sorts of processes are modeled in discrete states and and my my assumption is that that's a simplification because it makes work somehow more tractable right so it's manageable you can you can write the code on the basis of of this discrete state transitions for example um and and and still learn something even if you you sort of think that maybe state transitions are continuous for example right another thing that's more meta i suppose to that new is just it's sort of the following so we're quite inspired by a philosopher by the name of michael weisberg so his name is appears frequently now in the paper and he has this it's really this really nice paper from 2006 i believe it is all seven in biology and philosophy where he gives you two different approaches to a model based science we follow one in part because we don't think it's

very feasible to do the other one he refers to as a kind of brute force approach now i suppose this is going to give you all the high dimensionality that you can imagine that given computational power to pull it off on that approach the goal is to have the values of all the variables you can identify in your target system reflected in your model this is you have a super high degree if you like the correspondence between your model system and your target system the catch is that that particular ambition is extremely difficult given the sheer complexity of real world systems so the alternative is to say well then any other kind of model based approach in sciences if you can't achieve that degree of completeness if you like is to use idealizations so by necessity almost you have to simplify i suppose that would be one way of thinking about why we why we have low dimensional models and why one uses low dimensional models can you use experience to improve those models yeah i i'm not sure perhaps it seems like you ought to be able to ask a slightly different question but i see stephen wants to come back so i'll hold my comment for a minute yeah cool well just one that i think i suppose the dimensionality is so low that it's almost like even if you made it a thousand times more it still would be it's still useful because it shows the dynamics shifting which is a little bit different to most models however yeah there's like it's it's it's it's i don't know how many zeros it has been magnified by in terms of multiplying the complexity but that that that that um that might be anyway so that that i think it's a good point you're making though yeah thanks well let's take the casper hes deeply thought effect model i don't know if you know that one so there's a there's a variable in that model a matrix the g matrix which is which is trying to capture the minimization of expected free energy and the role that that plays in selecting between action policies and then and the idea is that uh that that g matrix is something that's changing over time such that you can do better or worse than expected at minimizing expected free energy so there's a there's a variable there which i guess is still there gonna yield models which are low dimensional in your sense but but it's still um being used to model something um that is i think phenomenological in the end namely valence the mole i would draw from that is that something that's an idealization an because it's low dimensional doesn't necessarily mean that that that modeling tool that you're using there that's that has this low dimensionality you can't still be used to describe something high level and phenomenological namely valence and and that goes back to what dave was saying before i think that the valence is actually something that's built into free energy minimizing systems because variational free energy a way of describing a way of indirectly representing is something that's fundamental to life and which is this tendency to avoid decay or so the low dimensionality i think doesn't uh doesn't

stand in the way of using this as uh to model very high dimensional phenomenologically deep thick concepts like valence yeah that's a really good example actually just one point on that and it also we're talking about general because i do agree with the idea of if we're saying what are generally tractable dynamical processes which maybe is another type of realism in a way or it's realism but to think of that it does show that affect becomes a plausible way which pretty well hasn't i don't know any other way to think about it previously it's why we start to get into consciousness as well with mark soames yeah it gives this plausible way where what was relegated in our scientific world because basically consciousness was taken away by psychology and feelings were kind of these this lower down ideas like well wait a sec this is actually the all-encompassing way to know what you think so that's actually because i actually do a lot of work with metaphors and one of the things that happens with metaphors is people what's the metaphor of knowing when you get something right like what's it like to know something so if you ask a mathematician what the answer is to a problem they don't get an answer on a computer screen they actually feel the answer being right so they've kind of somehow and then what's that like and then you can process that as a spatial metaphor in the room but the that you said it's affective so yeah i agree with what you're saying and i i what i'm interested as well is working backwards from an effective state and trying to plot out people's low dimensionality another way but that's for another conversation but i i i do agree with what you're saying but i don't know whether you would say dynamics is different to realism physical physical ways so in that model that the g vector is uh it's being used to set precision on action policy so which action policies can you be confident in that's a kind of metacognitive process in that it's not just acting based on some gut feeling you know in in the way that you described but but using that feedback from the body to to uh assign precision so to to look at how reliable a prediction error is in relation to your predictions when the quantity that you're minimizing is this expected free energy quantity um so there's something global going on there i think which gets you closer to what you what you're looking for when you're thinking about consciousness something in the global dynamics when you're thinking about the g vector i'll just read out one question from the chat and the answer could be as simple as read the paper or give a thought and then i'll ask a closing question for the authors so the question is from matthew mctigue and they wrote how do we resolve the controversy of the conflation between heuristic and markov what metaphysical legwork should be done to defend an ontological markov blanket okay so so the question is something like we don't really take that opportunity one sneaky movement they say go go read the greenback idiot paper but

moving from heuristics to ontology i think i think julian has already touched on that quite a bit when speaking on the sort of multi-scale characteristics of self-organizing um such that you just you so if you have as part of your model and this is our preferred approach here uh the formula isn't like the mark of blanket formulism then of course you're not going to be saying that heuristic this would be the literalist fallacy it's going to just be mappable onto this multi-scale structure that comprises the system so now you have to interpret that's the next step in this modeling move you have to interpret these sorts of notations under a certain description such that you can start thinking about what it is about biological uh organization that reflects in that indirect sense that we're working with these sorts of heuristics so that's the work that one should do but it's not it's not the kind of thing that we discuss at length in one section of the paper we call the section the paper moves across four different domains that has to be done with some swiftness and level of generality to bring the philosophical point home that would i mean i think that that's that's a separate paper actually that that little question so thank you matthew for that so just one closing um question as you are reviewing and revising the who is your audience and how would it impact the next several months or years of active inference and free energy principle research and application if people were to really direct the regime of attention and understand what you're talking about and update their generative models accordingly william do you want me or usually go first i'm happy to go first initially with initial thoughts so i wanted to to bring up recent paper by miguel aguileria and um chris buckley and others um where the problems that they raised there about how to take the free energy and uh and use it to to model real biological cognitive systems and they take very simple system and then show what kinds of assumptions you need to build into the free energy principle in order to use it to model those systems and they argue that those assumptions are are unrealistic and they pick out a very small class of systems and those systems are unlikely to be biological cognitive systems so that looks like bad news for the free but not from the perspective of our because what our paper shows is that all of those assumptions that they identify are actually exactly the kinds of idealizations and approximations that that don't rule out taking the free energy principle to indirectly represent biological cognitive systems and so the contribution of our paper to the community i think is to just point out that you know even if we look at the assumptions that underlie the free energy principle and think well those don't literally apply to that doesn't rule out the possibility of still making active inference models and and using them to to learn important things about their biological cognitive systems it would only rule that out if you think those systems literally have to instantiate the assumptions

that that are built into the fep before you can apply the fep and but we've argued no those assumptions don't need to literally obtain and they can just be idealizations and so so the contribution i think from of our paper to the community is to say there are these real challenges but they don't stand in the way of using the free energy principle to carry on doing active inference models and learn something important from those models okay now if we can achieve that that's certainly a job well done um i think in terms of the terms of the audience perhaps outside um just the technical very specific niche of the the free energy principle but i mean one one would like to hope that a bit like when you had steven uh on a few weeks back talking about free energy a user's guide which we crafted as a sort of the ambition in part was of course to introduce these sorts of conversations into the context of philosophy of biology and then hopefully they'll they will they will see the light of day i think similarly here and i'm sure julian will agree that you know i think these these sorts of papers that take up like broad picture discussions and the philosophy of science with say an emphasis on the free energy principle will bring an audience from the philosophy of science into contact with with the kind of you know area speciality that we're working on i would like to see that as a as an aim as well i think secondly as well or secondly just that just working through with some precision these kinds of conceptual details in this highly mathematical and and sometimes really abstract domain that is the free energy principle is important precisely also i think for the sciences working on it to have a feel for what what is the conceptual assumptions of making some claim right so so you're also like conceptually sort of engineering or this this this debating in a particular sense that can be fruitful i think for non-philosophers working on a knife a bit like what julian was touching on awesome well michael and julian thanks so much for the paper and also ian for co-authoring on the paper and best of luck with it dave dean and steven and also matthew thanks for participating this is a really interesting discussion and we hope to continue the conversation however you all would like thank you thank you thanks guys good night see you later everybody you